

THE ADVERSARIAL REASONING METHOD

A Conceptual Framework Proposal Emphasizing Reflection and Long-Term
Pattern Integration for Structured Clinical Diagnostic Reasoning

Not Yet Empirically Validated

Proposed for Pilot Evaluation

Working Draft & White Paper

Claudia West

[westcw01@gmail.com]

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Abstract

Diagnostic error remains among the most widespread and consequential sources of preventable harm in modern medicine. The relationship between diagnostic error and its causes is not very well captured by a binary split between “knowledge failures” and “reasoning failures.” A substantial body of evidence indicates that diagnostic accuracy rests primarily on the availability of relevant knowledge in memory rather than on the reasoning strategy applied to it (Schmidt & Mamede, 2015; Norman et al., 2024; Pelaccia et al., 2025). This paper does not treat the reasoning process and knowledge as competing explanations for error, rather, it treats them as interacting: reasoning failures most often surface as incomplete/poorly retrieved knowledge, and structured reasoning support is best understood as a way of helping clinicians deploy the knowledge they already have more completely and more self-critically, not as a substitute for acquiring it.

This paper proposes the Adversarial Reasoning Method (ARM), an eight-step framework for clinical diagnostic reasoning organized around the steps Observe, Represent, Compete, Attack, Challenge, Update, Reflect, and Integrate. Five of these steps are, by the paper’s own account, procedural restatements of previously published work (specifically Bordage, Kassirer, and Croskerry). ARM’s actual claim to contribution is narrower and is stated as such throughout: sequencing those elements into one connected procedure, converting a situational debiasing tool into a mandatory step, and closing the loop with explicit Reflect and Integrate stages; extending the tradition of *deliberate reflection* (Mamede, Schmidt, & Rikers, 2007; Mamede & Schmidt) into a fixed position within a whole-case sequence rather than proposing it as an independent intervention.

The framework is presented explicitly as a conceptual proposal. It has not been empirically tested, and no claim of superiority over existing models of clinical reasoning is made. This paper also proposes a small-scale case-series pilot as a feasibility step preceding any larger validation study, and addresses two practical concerns often absent from clinical-reasoning frameworks: the conflict between procedural rigor, clinical throughput, and case volume; and the operational constraints of introducing structured adversarial documentation into real clinical and educational settings.

(The intended audience for this document is clinician educators, program directors, and medical-education researchers who might sponsor, critique, or help design a pilot evaluation. It is not intended as a finished curriculum, a validated instrument, or a claim of practice-ready status.)

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1. The Problem: Why Diagnostic Errors Occur

1.1 Scope

Diagnosis is among the most fundamental responsibilities in medicine. Diagnostic errors can lead to delayed treatment, inappropriate intervention, avoidable cost, and patient harm.

In the United States, approximately 12 million adults experience a diagnostic error in the outpatient setting annually (Singh et al., 2014). Autopsy studies have consistently demonstrated a 10 to 40 percent discordance rate between clinical and post-mortem diagnoses, a finding that has remained largely stable for decades despite advances in imaging, laboratory medicine, and information technology (Hoppe et al., 2025). The National Academy of Medicine has identified improving diagnosis as a critical health-system priority (National Academies of Sciences, Engineering, and Medicine, 2015).

1.2 Cognitive Sources of Diagnostic Error, and Their Relationship to Knowledge

Diagnostic reasoning involves both intuitive pattern recognition and analytical evaluation. Errors often arise when rapid, intuitive judgements are not properly checked against new or disconfirming information. The following biases are among the most frequently implicated:

- Anchoring bias: excessive commitment to the initial diagnostic impression
- Premature closure: accepting a diagnosis before sufficient evidence has been gathered
- Confirmation bias: selectively seeking evidence that supports an existing hypothesis
- Availability bias: overestimating the probability of vivid or recently encountered diagnoses
- Framing effects: allowing referral source or case narrative to constrain the diagnostic search
- Base-rate neglect: failing to weight prior probability appropriately

A substantial and consistent line of research complicates any simple attribution of these errors to the “reasoning process” as distinct from knowledge. Elstein, Shulman, and Sprafka’s foundational work (1978) and its more recent extensions argue that diagnostic accuracy rests more on retrieval of relevant knowledge from memory than on the problem-solving strategy used to process it: it is not primarily the cognitive processes that are at fault in diagnostic error, but limitations in the knowledge that powers them (Pelaccia et al., 2025; Elstein, Shulman, & Sprafka, 1978). Pattern recognition has been shown to be no less vulnerable to diagnostic error than analytical reasoning, and additional deliberation time does not reliably correct bias or error on its own (Schmidt & Mamede, 2015; Sherbino et al., 2014; Sibbald et al., 2019). Multiple studies converge on the conclusion that diagnostic errors are mostly attributable to knowledge deficiencies or gaps rather than to a faulty reasoning procedure per se (Audétat et al., 2013; Norman, Sherbino, et al., 2014; Sibbald et al., 2019; ALQahtani et al., 2016; Monteiro et al., 2015, 2024).

This paper takes that literature as its starting assumption rather than as an objection to be answered after the fact. ARM is not framed here as evidence that “reasoning” rather than “knowledge” causes diagnostic error, as that framing is rejected as a false dichotomy. Norman and colleagues put the strongest version of the knowledge-first position directly: error is not principally a function of bias or incorrect system function, but an absence of the knowledge required for a correct solution, whether that means the clinician lacks a specific prior solution (experimental knowledge) or lacks a sufficient formal algorithm to construct a novel one (formal knowledge) (Norman et al., 2024). ARM’s premise is narrower, and, this paper argues, compatible with that position: even where the relevant knowledge is present, retrieval and deployment of it under time pressure, incomplete data, and premature narrative closure can still fail. ARM

is offered as a scaffold for using existing knowledge more completely and more self-critically, not as a mechanism that manufactures knowledge a clinician does not have, and not as a claim that reasoning process is the dominant driver of error relative to knowledge.

1.3 Reasoning Under Uncertainty

Clinicians routinely act on incomplete histories, evolving presentations, and imperfect tests. Effective diagnostic reasoning depends on managing that uncertainty without converging prematurely on a single explanation, which itself depends on having enough retrievable knowledge to recognize when convergence is premature.

1.4 The Social and System-Level Contribution

Cognitive and knowledge factors alone do not fully explain diagnostic error. A growing body of work situates clinical reasoning as socially and situationally distributed rather than purely intrapsychic: reasoning occurs within teams, handoffs, hierarchies, and institutional workflows that shape what information is valuable, which hypotheses get voiced, and whose disconfirming judgement is taken seriously (Koufidis et al., 2021, 2022). Time constraints, throughput pressure, EHR burden, weak diagnostic feedback loops, and hierarchical communication failures compound individual clinician limitations, obscuring what a clinician was in fact capable of. This paper treats these social and situational factors as part of the core conceptual model, not as background context: Sections 4.1, 9, and 10 return to them directly, and the Reflect and Integrate steps (Sections 5, 8) are framed throughout as socially embedded practices conducted with colleagues, supervisors, or a case-review setting, rather than solitary cognitive exercises.

1.5 Problem Statement

Existing education approaches provide strong instruction in clinical knowledge and differential diagnosis, but fewer frameworks explicitly walk a clinician through structured hypothesis generation, self-critique, uncertainty monitoring, and post-case reflection as a single connected sequence; fewer still extend that sequence into the socially embedded, longer-arc practice of deliberate reflection after the case is closed. This paper proposes such a sequence, is explicit about which parts of it are new and which are not, and, per Section 3.4, locates its actual contribution primarily in the later stages of that sequence rather than the earlier ones.

2. Existing Approaches and Their Limitations

2.1 Illness Script Theory

Illness scripts encode the enabling conditions, mechanisms, and characteristic findings of a disease. Expert clinicians build rich illness scripts through repeated exposure, which enables fast, often accurate pattern matching. The theory is primarily descriptive: it explains expert retrospective reasoning, but offers no normative procedure for novices, atypical presentations, or systemic bias detection.

2.2 Hypothetico-Deductive Reasoning

Kassirer’s hypothetico-deductive model describes generation of a small set of candidate diagnoses early in the encounter, followed by targeted information gathering to confirm or refute each one. It has strong empirical support across levels of expertise. Its main limitation is the absence of a formal mechanism for detecting when a hypothesis has been inadequately challenged; clinicians frequently gather confirming evidence without systematically seeking disconfirming data, an error the model does not structurally prevent.

2.3 Differential Diagnosis Approaches

Traditional differential diagnosis generates and progressively narrows multiple candidate explanations. It guards against fixation on a single diagnosis but offers limited guidance on how competing diagnoses should be challenged, prioritized, or retrospectively evaluated after resolution.

2.4 Bayesian Reasoning

Bayesian approaches formalize updating of diagnostic probability using prior probability and likelihood ratios. They provide a mathematically rigorous account of belief revision, but formal real-time computation is rarely feasible: clinicians lack precise likelihood ratios for most findings, priors are highly context-dependent, and tracking more than a few hypotheses numerically exceeds practical cognitive capacity. In most clinical settings, Bayesian reasoning functions as a conceptual orientation, not a real-time calculation.

2.5 Dual-Process Theory Interventions

Educational interventions built on dual-process theory, including Croskerry’s cognitive forcing strategies, prompt clinicians to pause and apply deliberate analysis before finalizing a diagnosis. Evidence for sustained real-world effectiveness is mixed. The core limitation is structural: metacognitive awareness without a concrete procedural scaffold produces inconsistent results. Clinicians may recognize the concept of bias without a repeatable method for identifying and correcting it mid-reasoning.

2.6 Why These Approaches Remain Complementary Rather Than Competitive

ARM is not proposed as a replacement for illness scripts, Bayesian reasoning, differential diagnosis, or hypothetico-deductive reasoning. It is proposed as a procedural sequence within which these approaches can operate: illness scripts inform hypothesis generation, Bayesian principles can inform probability revision, and differential diagnosis defines the competing possibilities under consideration. Section 3 makes this relationship explicit.

2.7 Deliberate Reflection (Mamede, Schmidt, and Colleagues)

A body of work associated primarily with Silvia Mamede and colleagues, coined *deliberate reflection*, proposes a structured intervention in which clinicians systematically compare and contrast findings

against competing diagnoses, generate expectations for each, and check those expectations against the case data, with demonstrated effects on diagnostic accuracy for complex cases under controlled conditions (Mamede, Schmidt, & Rikers, 2007; Schmidt & Mamede, 2015). This literature offers an intervention that substantially overlaps with what ARM proposes across its early-to-middle steps (Sections 5, Steps 1-4): systematic comparison of hypotheses against evidence, explicit generation of disconfirming expectations, and resistance to premature closure.

This paper does not present ARM's early steps as an independent invention alongside deliberate reflection. ARM is positioned here as one way of organizing, sequencing, and extending deliberate reflection; embedding it as a mandatory checkpoint within a larger procedure that also specifies what happens after it (Reflect and Integrate, Sections 5 and 8), which is not deliberate reflection's primary focus. Readers seeking the empirical grounding for the disconfirmation-under-uncertainty mechanism ARM's early steps rely on should treat the deliberate reflection literature, not this paper, as the evidentiary source; this paper's own contribution in that region is procedural sequencing, not independent validation.

2.8 Prospective and Retrospective Models of Reasoning (Complementary, Not Competing)

ARM is a prospective framework: it structures reasoning as a case unfolds, before an outcome is known. This is a different orientation from retrospective cognitive-error-analysis frameworks, such as the root-cause-analysis-based curriculum described in *Rooted in Reasoning* (Klimpl & Staudinger, 2025), which teaches clinicians to analyze reasoning failures after a diagnostic error has already occurred. That curriculum is itself instructive on this point: its own four modules move from prospective foundations (problem representation and illness scripts; dual-process theory, cognitive bias, and bias mitigation; probabilistic and Bayesian reasoning) to a fourth and final module of retrospective case analysis, built explicitly on the prospective grounding the first three modules establish. ARM and a retrospective curriculum of this kind are therefore complementary rather than competing: retrospective analysis of a cognitive error can only be meaningful once a clinician has some baseline understanding of how they are expected to reason and how that reasoning predictably breaks down under specific conditions, an understanding a prospective framework is intended to build in the first place. ARM does not propose to replace retrospective error analysis; it is offered as a prospective counterpart that such retrospective work presupposes, in much the same relationship Klimpl and Staudinger's own curriculum structures between its earlier and later modules.

2.9 Related Teaching and Assessment Tools: SNAPPS and ART

2.9.1 SNAPPS: A Case-Presentation Structure

SNAPPS (Summarize, Narrow, Analyze, Probe, Plan, Select) is a six-step, learner-centered model for structuring how a learner presents a case to a preceptor in outpatient teaching (Wolpaw, Wolpaw, & Papp, 2003). The learner summarizes the history and findings, narrows the differential to two or three live possibilities, analyzes the differential by comparing and contrasting them, probes the preceptor with questions about uncertainties or alternative approaches, plans management, and selects a case-related topic for self-directed learning. The Probe step in particular explicitly invites the learner to voice uncertainty rather than project unearned confidence, a goal ARM shares, most directly with Step 2 (Represent) and Step 4 (Attack). SNAPPS, however, is a presentation structure for a single live encounter with a supervisor: it does not specify a formal adversarial step against the leading hypothesis, does not extend past the encounter into structured post-case reflection, and has no equivalent to ARM's Integrate stage.

2.9.2 ART: A Reasoning Assessment Instrument

The Assessment of Reasoning Tool (ART) is a behaviorally anchored, three-point rating scale developed by the Society to Improve Diagnosis in Medicine for faculty to assess a learner’s clinical reasoning during an oral case presentation, typically one structured using a model like SNAPPS (Thammasitboon et al., 2018). ART rates five domains: hypothesis-directed data gathering, articulation of a problem representation, formulation of a prioritized differential diagnosis, diagnostic testing aligned with high-value-care principles, and metacognition. ART is not itself a reasoning procedure; it is a measurement instrument applied to reasoning a learner has already performed and presented. It therefore does not compete with ARM at the level of method, it operates one level up, as an evaluation layer that could, in principle, be applied to reasoning produced under ARM just as it can to reasoning produced under any other model or none at all.

2.9.3 Honest Statement of Contribution (Relative to SNAPPS and ART)

SNAPPS and ART operate at different levels than ARM: SNAPPS structures how a case is presented in the moment, and ART measures the quality of reasoning already demonstrated. Neither proposes a structured adversarial step against a leading hypothesis, and neither extends into post-case reflection or long-term pattern integration the way ARM’s Reflect and Integrate stages do. Where this paper sees a genuinely useful connection, rather than a competitive one, is instrumental rather than conceptual: ART’s five domains and behaviorally anchored scale could plausibly serve as an off-the-shelf outcome measure for a future ARM pilot (Appendix B) rather than requiring a bespoke instrument built from scratch. This is a modest, practical connection, not a claim that ARM subsumes or improves on either tool. A fuller treatment of overlap and difference, of the kind given to Bordage, Kassirer, Croskerry, and Mamede/Schmidt in Section 3, remains a reasonable next step for future revision rather than something this section claims to have completed.

3. Positioning Relative to Bordage, Kassirer, and Croskerry

Readers familiar with the clinical reasoning literature will recognize substantial overlap between ARM's steps and three specific published models: Bordage's work on problem representation and semantic qualifiers, Kassirer's hypothetico-deductive method, and Croskerry's cognitive forcing strategies — as well as, per Section 2.7, Mamede and Schmidt's deliberate reflection literature. This section states that overlap directly rather than leaving it to be discovered, and separates what is adopted from what is claimed as new.

3.1 Bordage: Problem Representation as an Externalization Tool

Bordage's account of problem representation, compressing a case into a compact, semantically qualified summary (for example, framing findings by tempo, severity, and pattern rather than a raw symptom list), is the direct source of ARM's Step 2 (Represent). ARM adds no new theoretical content here, and this paper is explicit about a further limit: the model is better supported as correlational (experts who reason better also tend to encapsulate cases more efficiently) than as a causal mechanism that reliably improves reasoning when taught or mandated (cf. Bordage & Lemieux, 1991).

ARM therefore does not claim that mandating Step 2 improves diagnostic accuracy. Represent is retained in the sequence as an externalization and communication checkpoint: it forces a clinician to state, out loud or in writing, a compressed pattern-level summary before hypothesis generation begins, which is valuable for making reasoning legible to supervisors, teammates, and the clinician's own later self-review (Section 8, Section 2.7) independent of whether the act of compression itself measurably improves accuracy. This is a narrower and more defensible claim than the one made in earlier drafts of this paper.

3.2 Kassirer: Hypothetico-Deductive Reasoning

Kassirer's model of early hypothesis generation followed by targeted, confirmatory-or-refuting data gathering underlies ARM's Steps 1, 3, 5, and 6 (Observe, Compete, Challenge, Update) taken together. ARM does not introduce a new cognitive mechanism here. What differs is procedural: Kassirer's model does not itself mandate a distinct adversarial step against the leading hypothesis, and does not specify a formal post-hoc reflection or long-term integration stage. ARM's contribution in this region is sequencing and mandatory documentation, not new theory.

3.3 Croskerry: Cognitive Forcing Strategies

Croskerry's "consider the opposite" strategy and broader catalogue of cognitive forcing strategies are the direct antecedent of ARM's Step 4 (Attack). Croskerry presents disconfirmation-seeking as one situational tool among several bias-mitigation strategies, to be deployed at the clinician's discretion. ARM's difference is that it makes this step non-optional and places it at a fixed point in a whole-case sequence, paired with a structured set of forced questions (what evidence would exist if I am correct; if I am wrong; what would a disagreeing colleague argue) rather than a general reminder to consider alternatives.

3.4 Honest Statement of Contribution

Taken together, ARM's original contribution is limited to three things: sequencing previously separate patterns, including the deliberate reflection literature, into one ordered procedure; converting Croskerry's optional debiasing tool into a mandatory, fixed-position step; and closing the loop with explicit Reflect and Integrate stages, which have no clear precedent across these four bodies of work. Steps 1, 2, 3, 5, and 6 are, in substance, procedural restatements of Bordage's, Kassirer's, and Mamede and Schmidt's published work. Where this paper cannot identify a defensible claim to novel cognitive content, it says so

here rather than implying otherwise: ARM is best described as a synthesis and procedural sequencing of existing, separately published models, with the Attack step's mandatory placement and the Reflect/Integrate stages as its only structurally original elements. This paper's title and framing (see Abstract, Section 13) are oriented around that narrower claim rather than around the eight steps as a whole.

4. The Adversarial Reasoning Method (ARM): Framework Overview

4.1 Design Principles

ARM rests on five premises:

- Diagnostic accuracy depends primarily on the availability and retrieval of relevant knowledge; structured reasoning support is offered as a way to deploy that knowledge more completely and self-critically, not as a substitute for it (Section 1.2).
- Hypothesis testing requires deliberate adversarial effort; disproving one's own leading diagnosis is a different cognitive act than passively accumulating supporting evidence.
- Clinical intuition is compressed expertise, not guesswork, and should be trained and interrogated rather than suppressed or blindly trusted.
- Diagnostic certainty is provisional; effective reasoning requires continuous reassessment of confidence and willingness to revise favored explanations.
- Reasoning is socially and situationally embedded, not purely intrapsychic; hierarchy, workflow, and team communication shape whether correct reasoning surfaces and is acted upon (Section 1.4).

The framework is intended to be used stepwise in high-stakes or undifferentiated cases and selectively compressed in routine ones (see Section 9). It is a cognitive scaffold, not a mandatory protocol for every encounter.

4.2 The Eight Steps at a Glance

#	Step	Core Function	Cognitive Failure Addressed
1	Observe	Structured intake of clinical/contextual data; separation of observation from interpretation	Incomplete data gathering
2	Represent	Compression of findings into a concise, pattern-oriented summary (externalization tool)	Diagnostic ambiguity
3	Compete	Generation of competing hypotheses with supporting/contradicting evidence	Premature closure
4	Attack	Adversarial audit of the leading hypothesis via structured disconfirming questions	Confirmation bias, anchoring

5	Challenge	Test selection based on expected impact on the leading decision, not breadth of hypotheses moved	Low-value testing
6	Update	Iterative revision as new data accumulate	Diagnostic inertia
7	Reflect	Structured, socially embedded post-case self-evaluation	Failure to learn
8	Integrate	Long-term pattern formation across cases	Passive experience accumulation

Although presented sequentially for instructional clarity, movement between steps is iterative. New information may prompt renewed observation, reformulated problem representation, or repeated adversarial testing.

5. Step-by-Step Explanation

Step 1: Observe: Case Intake and Symptom Inventory

- Document age, sex, occupation, environmental exposures, symptom timeline, past medical history, medications, allergies, lifestyle factors, and psychosocial context.
- Separate positive findings from negatives; absence is often as diagnostically significant as presence.
- Distinguish observation from interpretation: “patient reports chest pain” is data; “cardiac chest pain” is a hypothesis. These must not be conflated at this stage.

Step 2: Represent: Problem Representation

- Use age, tempo, and system involvement as the primary framing elements (Bordage).
- Transform a symptom list into a diagnostic pattern, e.g. “subacute multisystem inflammatory illness in a previously healthy middle-aged man.”
- Avoid premature diagnostic language; “multisystem inflammatory process,” not “lupus.”
- Treat this step as externalization for communication and later self-review, not as a step assumed on its own to improve accuracy (Section 3.1).

Step 3: Compete: Hypothesis Generation

- For each hypothesis, document supporting evidence, contradicting evidence, relative likelihood, and consequences if missed.
- Document intuitive impressions separately from formal reasoning; they are a distinct cognitive signal worth interrogating, not a substitute for evidence.
- Likelihood is expressed comparatively by default (“more likely than,” “roughly equally likely”). See Section 6 for the rationale and an optional numerical extension.
- Hypothesis count should scale with uncertainty: one or two for a clear presentation, five or more for an undifferentiated one.

Step 4: Attack: Structured Disconfirmation

- For the leading hypothesis, ask: What evidence must exist if I am correct? If I am incorrect? What single finding would most challenge my model? What am I most likely overlooking? What would a disagreeing colleague argue?
- Identify the single best alternative hypothesis that accounts for the same data.
- Where feasible, perform this step with or in front of another clinician or learner; disconfirmation surfaced socially is more robust than disconfirmation attempted alone (Section 1.4).

Step 4 deliberately does not ask clinicians to ‘name the biases’ operating in a case. There are dozens of catalogued cognitive biases, and no good evidence that clinicians can reliably identify which one is influencing them in the moment, or that naming a bias category improves subsequent judgment (Norman et al., 2014; Norman et al., 2017; Sibbald et al., 2019; Zwaan et al., 2017). The disconfirming questions

above are retained instead because they target behavior directly, rather than requiring accurate bias self-diagnosis first.

Step 5: Challenge: Test Selection and Evidence Mapping

- Prioritize tests by their expected impact on the leading management decision, not by how many competing diagnoses a result could move. The highest-value test is often the one that decisively confirms or excludes a single high-consequence hypothesis; test value depends on what hinges on the answer, not on the number of diagnoses it touches.
- Weigh sensitivity, specificity, false-positive/negative rates, invasiveness, risk, and cost against how much a given result would actually change the differential or the plan — whether by settling one consequential hypothesis or by discriminating among several.
- Document the expected impact of each possible result on the differential before the result is known.

Step 6: Update: Iterative Revision

- Classify prior assumptions as correct, incorrect, or irrelevant in light of new data.
- Actively reconsider previously dismissed diagnoses. New data may rehabilitate a low-probability hypothesis.
- Avoid interpreting ambiguous new data as automatically confirming the existing impression; confidence should move up, down, or stay level based on the actual quality of new evidence.

Step 7: Reflect: Post-Case Analysis

- Ask: What did I observe correctly? What did I miss? What changed my model, and when?
- Identify near-misses: diagnoses almost missed, held too long, or investigated too late.
- Reflect regardless of outcome; a correct diagnosis can still result from flawed reasoning that will eventually fail.
- Conduct this step, where possible, as a shared exercise (with a supervisor, peer, or case-review group) rather than a purely private one, consistent with the socially embedded model in Section 1.4.

Step 8: Integrate: Pattern Formation and Long Term Learning

- Compare cases over time to identify presentation patterns and recurring cognitive vulnerabilities.
- Distinguish genuine pattern recognition from narrative coherence; a case “making sense” in hindsight does not validate the process that reached the conclusion.
- Intuition developed through systematic reflection may be more reliable than intuition from passive experience alone.

6. Optional Numerical Extension For Advanced Use

6.1 Why Comparative Language Is the Default

Steps 3 and 6 use comparative likelihood language (“more likely than,” “much less likely unless”) rather than numerical probability estimates by default, for three specific reasons rather than by omission:

- Cognitive load: assigning and tracking numerical priors and posteriors across several competing hypotheses in real time exceeds practical working-memory capacity for most clinicians, particularly under time pressure (see Section 2.4).
- Calibration error risk: untrained clinicians asked to produce numbers tend to produce false precision — a percentage that feels rigorous but is not actually derived from a likelihood ratio or a real prior. Overconfidence built on invented numbers is arguably worse than acknowledged qualitative uncertainty (Berner & Graber, 2008).
- Portability: comparative language transfers across specialties and experience levels without requiring shared numerical literacy or agreed-upon likelihood ratios, which rarely exist for most clinical findings.

6.2 Optional Extension for Trained or Research Users

For clinicians or research contexts with genuine training in Bayesian reasoning, an optional numerical layer can be applied at Steps 3 and 6 without changing the framework’s structure:

- At Step 3, assign a rough prior probability band to each hypothesis (e.g., <10%, 10–50%, >50%) rather than a false-precision point estimate, or a full prior where a defensible base rate exists.
- At Step 6, apply likelihood ratios where they are actually known (e.g., a well-characterized laboratory test) to compute a posterior; otherwise retain the qualitative default rather than creating a likelihood ratio.

This extension is opt-in and is not recommended as a default for trainees without prior instruction in Bayesian updating, for the calibration reasons stated in 6.1. A pilot evaluating ARM (Section 8) could usefully test whether the numerical extension improves or degrades reasoning quality relative to the comparative default among users with and without prior Bayesian training.

7. Worked Case

The following case demonstrates application of all eight steps to a common, instructive presentation. This is a single illustrative example, not evidence of the framework's effectiveness; Section 8 proposes how a small series of such cases could be used as a pilot.

Step	Application
1: Observe	19-year-old woman. Sore throat and fever for two days, feels very tired, swollen glands in the neck. No cough, no runny nose. Roommate had strep throat last week. Negative factors: no rash, no cough, no runny nose.
2: Represent	Sudden sore throat and swollen glands in a young adult, without the cough or runny nose that usually points to a common cold.
3: Compete	Leading: strep throat (a bacterial infection). Competing: A viral cold-type infection; mononucleosis (“mono”), which is common in this age group and also causes sore throat, fever, and swollen glands, but usually comes with more extreme tiredness. Strep is supported by the sudden fever, swollen glands, and no cough; mono is retained because of how tired the patient feels and their age.
4: Attack	Is strep too obvious a fit just because the roommate had it (availability bias risk)? What would prove this wrong; if the strep test is negative, or the fatigue is more severe than a sore throat alone would explain, mono becomes more likely and changes what the patient should be told to avoid.
5: Challenge	A rapid strep test (a simple throat swab with results in minutes) is done first, because a negative result would meaningfully point toward mono rather than just ruling out strep. If negative and fatigue becomes severe, a mono blood test is the next step.
6: Update	Rapid strep test comes back negative. Given the ongoing severe fatigue, a mono blood test is ordered and comes back positive. Revised diagnosis: mono, not strep throat.
7: Reflect	Availability bias was a real risk here; it was tempting to assume strep just because the roommate had it recently. The severity of the fatigue was the clue that it wasn't strep and was almost explained away by “just feeling sick.” Testing rather than assuming prevents a missed diagnosis.
8: Integrate	Pattern filed: sore throat + fever + swollen glands + fatigue that feels disproportionate to the sore throat, in a teenager or young adult, should raise mono as a real possibility, not just strep, especially when someone nearby recently had similar symptoms and it's tempting to assume their diagnosis in a slightly different presentation

Teaching point: the value in this case was not the diagnosis. Strep throat is readily identified. The value was catching mononucleosis instead, a diagnosis that changes what the patient needs to be told, and that would have been missed by assuming strep because it was “going around.” The framework forces documentation of reasoning rather than conclusions.

8. Proposed Case-Series Pilot

A single worked case, however clean, demonstrates only that the framework can be applied, not that it improves reasoning or is usable at scale. Before any randomized or simulation-cohort trial (Section 12), a small case-series pilot is proposed as a feasibility step.

8.1 Purpose

The pilot's purpose is feasibility and instrumentation, not efficacy. It is designed to answer operational questions that a larger trial would otherwise have to guess at:

- Do independent users of the framework converge on similar Attack-step outputs (leading hypothesis, best alternative, key disconfirming question) for the same case?
- How long does full documentation actually take, per step, across case types of varying complexity?
- Which steps compress naturally without loss, and which are skipped under time pressure even when the case warrants them?
- Does the framework surface a near-miss or reconsidered diagnosis that the rater's own initial, pre-ARM impression of the case would plausibly have missed? (This is a within-rater comparison against each rater's own snap impression, not a claim about unstructured reasoning in general; the design below has every rater apply ARM, so it cannot support a between-groups comparison against a separate unstructured-reasoning arm.)

8.2 Proposed Design

- Assemble 5–8 de-identified case vignettes spanning a range of acuity, specialty, and diagnostic difficulty (include at least one straightforward case and at least one genuinely ambiguous one).
- Recruit 2–4 raters of comparable training level (e.g., senior medical students or resident-level reviewers) to independently apply the full eight-step ARM documentation to each case.
- Before applying the ARM steps, ask each rater to record their initial, unstructured impression of the case — leading diagnosis and confidence — so the within-rater comparison in Section 8.1 has something to compare against.
- Time each step per rater per case.
- Score inter-rater agreement on: (a) identity of the leading hypothesis, (b) identity of the named best alternative, (c) whether a disconfirming test was proposed at all, and (d) whether a near-miss or previously dismissed diagnosis was reconsidered at Step 6.
- Collect brief qualitative feedback from raters on which steps felt redundant, unclear, or artificially forced.

8.3 What This Pilot Would and Would Not Show

A favorable result, reasonable agreement on best-alternative identification, manageable completion times, and raters reporting the Attack step changed their confidence, would justify investment in the larger designs described in Section 12. It would not, on its own, demonstrate reduced diagnostic error, since case vignettes are not real patients and small convenience samples of raters are not powered for anything

beyond feasibility. This distinction is stated explicitly so the pilot is not later mischaracterized as validation.

9. Rigor, Throughput, and Case Volume

The full eight-step sequence is deliberately System-2-heavy: it asks for explicit documentation at each stage, which takes real time. Real clinical environments are throughput-constrained, and a framework that ignores that conflict is not usable, whatever its theoretical merit.

9.1 Two Distinct Costs, Not One

Time spent on structured documentation for one case is time not spent seeing additional cases, and case volume itself is the primary mechanism by which System-1 pattern recognition and illness-script density are built (Section 2.1). A framework that measurably improves reasoning quality per case but reduces the number of cases a trainee sees over a rotation could plausibly produce a net negative effect on long-run diagnostic competence, an outcome ARM's own logic would have to treat as a failure. This tradeoff is treated here as a distinct empirical question from documentation burden.

9.2 Reframing Tiering as a Research Question

Rather than prescribing fixed thresholds, the table below is offered as a heuristic starting point for a pilot, not as a validated rule. The governing research question this paper proposes is: under what circumstances do clinicians find more systematic diagnostic-reasoning strategies worth their time and cognitive cost, and what contextual factors (case complexity, stakes, fatigue, supervision, institutional incentive) predict when clinicians deploy versus abandon a strategy they themselves would consider best practice? Section 9.1's volume-versus-depth tradeoff and the tiering question below are both instances of that same broader question, and are proposed as a dedicated line of pilot inquiry (Appendix B) rather than settled by assertion in this paper.

Tier	Trigger	Recommended Depth
Tier 1: Routine	Clear presentation, high pretest probability, low stakes if wrong	Observe > Represent only, applied informally without documentation
Tier 2: Moderate	Some diagnostic uncertainty or moderate consequence if wrong	Observe > Represent > Compete > Attack (the "minimum viable ARM"), briefly documented
Tier 3: High Stakes / Undifferentiated	Diagnostic uncertainty is high, consequences of error are serious, or presentation is atypical	Full eight-step sequence with documentation, including Reflect and Integrate after resolution

No data currently support these specific tier thresholds, the boundary between them, or the assumption that clinicians can accurately self-classify a case into the correct tier in real time. The case-series pilot in Section 8 could reasonably be extended to test whether raters correctly and consistently classify cases into these tiers, and Appendix B proposes a separate, smaller pilot addressing the volume-versus-depth question in Section 9.1 directly, before either is recommended for adoption.

10. Implementation Constraints

Beyond time cost, several practical constraints would need to be resolved before any pilot moves from paper to a real educational or clinical setting:

- Documentation burden and EHR integration: an ARM worksheet added on top of existing note-writing requirements competes for the same limited time as clinical documentation itself; any real deployment likely needs a lightweight, separate template rather than an addition to the permanent chart note.
- Documentation and disclosure of uncertainty: clearly documenting a differential and the reasoning behind a chosen course of action is not primarily a medico-legal liability. In practice, this kind of documentation is often valuable to the clinicians who subsequently care for the patient, since it makes the reasoning behind a decision legible rather than opaque. The more significant barrier is cultural rather than legal: clinicians, particularly junior clinicians, can be uncomfortable explicitly acknowledging what they do not know, and a framework that mandates this kind of disclosure needs to address that discomfort directly rather than treat documentation itself as the primary risk. Genuine medico-legal questions can still arise and are a matter for institutional and legal stakeholders to weigh in on, but they are not a reason to default to non-disclosure.
- Institutional and cultural buy-in: since clinical reasoning is not universally or explicitly taught in medical training, the more likely source of resistance is not that clinicians will feel scrutinized by mandatory self-disconfirmation exercises, but that experienced clinicians who have practiced without an explicit model like this for years or decades may not see why they would need to change now, having done well enough without it. A pilot needs at least one faculty champion willing to model the Attack and Reflect steps openly, and framing ARM as a scaffold that formalizes reasoning many clinicians already do implicitly, rather than as a corrective to a deficiency, may reduce this resistance more than addressing concerns about scrutiny alone.
- Training burden: the Attack and Reflect steps require intellectual honesty and tolerance for being shown wrong in front of peers or supervisors; without some baseline psychological safety, learners are likely to perform the steps superficially rather than genuinely.
- Scalability of rater/grader time: any pilot or curricular use that requires faculty review of ARM worksheets adds grading burden on top of existing teaching loads, which needs to be budgeted for explicitly rather than assumed to be free.
- Case-volume opportunity cost: any curricular deployment should track cases-seen-per-rotation alongside reasoning-quality measures, so that a gain in per-case rigor is not purchased with an undetected loss in case exposure (Section 9.1).

11. Strengths and Limitations

11.1 Strengths

- Procedural completeness: the eight steps span the full arc of a clinical encounter, from intake to long-term learning, in one integrated sequence.
- Mandatory adversarial step: making Croskerry’s “consider the opposite” strategy a fixed, non-optional step is a small but concrete procedural difference from treating it as one situational tool among many.
- Behavior-targeted disconfirmation: the Attack step pairs mandatory placement with specific, concrete interrogating questions rather than a general reminder to “watch for bias,” avoiding reliance on clinicians’ ability to self-diagnose which named bias is operating (Section 5, Step 4).
- Compatibility with existing models: the framework is explicit that it organizes, rather than replaces, illness scripts, Bayesian updating, hypothetico-deductive reasoning, and deliberate reflection (Sections 2.7, 3).
- Education tractability: each step has a defined purpose and a documentable output, which lends itself to teaching and to the pilot design in Section 8.
- Its most defensible original contribution (the Reflect and Integrate stages) targets a stage of the reasoning cycle (post-case, longitudinal) that is comparatively underserved by Bordage, Kassirer, Croskerry, and the deliberate reflection literature alike (Section 3.4).

11.2 Limitations

- No empirical validation: no prospective, simulation-based, or randomized data currently exist showing that ARM reduces diagnostic error. This is a structural gap, not a nuance, and this paper makes no claim to the contrary.
- A foundational objection from within the field: at least one senior clinical-reasoning theorist consulted on this paper does not consider procedural approaches such as hypothetico-deductive reasoning and cognitive forcing to be well-supported explanatory concepts for how clinical reasoning actually works, independent of any specific implementation. This is not a limitation of ARM’s execution; it is a live, unresolved disagreement about whether the procedural framing this entire paper relies on is the right level of description at all (a reviewer, personal communication, 2026). This paper does not resolve that disagreement and states it here rather than eliding it.
- Full completion of all eight steps adds real overhead; Section 9 proposes a tiering scheme and a distinct volume-versus-depth question to address this, but neither the thresholds nor the net effect on case exposure are tested.
- Quantification of uncertainty: the default framework is qualitative. Section 6 addresses this directly.
- Implementation friction: documentation burden, medico-legal exposure, and institutional buy-in (Section 10) are real barriers to adoption beyond a controlled pilot setting.
- Learning curve: the Attack and Reflect steps require tolerance for uncertainty, which may take training and a supportive environment to properly develop.

12. Future Development

12.1 Sequencing

This paper proposes the following sequence, each stage gated on the success of the one before it:

- Stage 1 (this paper): conceptual specification and internal consistency review.
- Stage 2: the case-series feasibility pilot described in Section 8, plus the volume-versus-depth pilot proposed in Appendix B.
- Stage 3: a simulation-based cohort study, in which medical students or residents apply ARM to standardized vignettes with pre-post assessment of diagnostic accuracy and reasoning-quality scores.
- Stage 4: a randomized controlled trial comparing diagnostic accuracy and error rates with a primary outcome of diagnostic error reduction as adjudicated by blinded expert panel review.

Moving to Stage 3 and/or 4 without a successful Stage 2 would mean testing an instrument whose basic usability and inter-rater consistency have not been established, which is a design flaw this roadmap is intended to prevent.

12.2 Integration With Medical Education (Longer-Term, Contingent on Validation)

- Pre-clinical: structured Compete/Attack drills in case-based learning, based on Stage 2/3 results.
- Clinical clerkships: a documentation tool for case presentations requiring explicit Represent and Attack steps.
- Residency: a structured Reflect/Integrate tool for morbidity and mortality conferences as an alternative to retrospective blame-oriented review.

12.3 Digital Integration (Speculative)

A structured ARM template embedded in an EHR or a standalone educational tool could, in principle, prompt documentation of competing hypotheses at the time of ordering and flag cases with no disconfirming test ordered against the leading hypothesis. This is listed as a long-term, speculative direction, not a near-term deliverable, and depends entirely on the medico-legal and workflow questions raised in Section 10 being resolved first.

12.4 Subspecialty Adaptation (Speculative)

- Emergency medicine: a compressed Observe > Represent > Attack sequence suited to a high-acuity setting.
- Radiology: Attack/Update sequences adapted to image interpretation rather than clinical synthesis.
- Pathology/laboratory medicine: Compete/Attack logic applied to laboratory interpretation.

13. Conclusion

Diagnostic error is not an unsolvable problem, but it will not be solved by reasoning-process reform alone, any more than by medical knowledge alone; the two interact, and the evidence weighs knowledge retrieval as the larger factor (Section 1.2). The Adversarial Reasoning Method (ARM) proposes a structured, sequenced procedure for clinical diagnostic reasoning, built largely by organizing and making mandatory a small number of previously published ideas: Bordage’s problem representation, Kassirer’s hypothetico-deductive testing, Croskerry’s cognitive forcing strategies, and Mamede and Schmidt’s deliberate reflection, and closing the loop with explicit reflection and pattern integration that this literature has not, on its own, formalized.

This paper makes no claim that ARM has been shown to reduce diagnostic error, and no claim of novelty beyond what Section 3.4 states plainly: sequencing, mandatory placement of the adversarial step, and the Reflect/Integrate stages. It also does not resolve the more fundamental disagreement, raised directly by a reviewer of this work, over whether procedural reasoning frameworks are the right unit of analysis for clinical reasoning at all (Section 11.2). The framework is offered here as a conceptual proposal, together with a concrete, appropriately modest pilot design (Section 8, Appendix B) intended to generate the feasibility data needed before any larger claim could be responsibly made.

The goal of this document is not to present a completely original framework. The goal is a testable proposal, honestly scoped, that could plausibly contribute to fewer missed diagnoses if it survives the pilot stages outlined above, and an honest account of why it might not.

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Appendix A: Changelog

Pre-v1.6 (Prior to July 2026)

The versions preceding v1.6 were merely conceptual development drafts rather than formal manuscript revisions. They established and iteratively refined the underlying architecture of ARM before the framework was circulated for substantive external review.

- Developed the initial concept of a structured clinical reasoning framework intended to reduce diagnostic error by making diagnostic reasoning more explicit, adversarial, and revisable.
- Developed and iterated the core ARM sequence, including observation, representation, hypothesis competition, disconfirmation/challenge, updating, reflection, and integration.
- Developed the concept of deliberately treating an initial diagnostic impression as provisional and subjecting it to structured attempts at disconfirmation.
- Developed the role of intuition as an informational signal rather than as either inherently reliable or inherently erroneous.
- Developed the distinction between the immediate diagnostic process and later reflective/integrative evaluation of that process.
- Iteratively refined the framework’s terminology, organization, and proposed use as a practical reasoning method.
- Expanded the framework from an individual reasoning concept into a proposed model for clinical reasoning, diagnostic error research, and future empirical testing.
- Produced the first substantially complete manuscript/white-paper form of ARM, which subsequently became the basis for v1.6.
- v1.6 was the first version treated as a formal manuscript revision and circulated for external feedback; subsequent versions therefore document substantive revision in response to that feedback.

v1.6 (July 2026)

- Circulated for external feedback from clinician-educators and clinical-reasoning researchers. Superseded by the changes below.

v1.7 (September 2026)

- Reframed the abstract, Section 1.2, and Section 13 around a knowledge/reasoning interaction model rather than a reasoning-vs-knowledge dichotomy; added citations to the knowledge-deficit literature raised in external feedback (Pelaccia, Norman, Schmidt & Mamede, Sherbino, Sibbald, Audetat, ALQahtani, Monteiro).
- Added Section 2.7, integrating Mamede and Schmidt’s deliberate reflection literature explicitly; repositioned ARM’s early steps as organizing/extending that literature rather than independently inventing structured self-disconfirmation.
- Removed “Name the biases” instruction from Step 4; retained the behavior-targeted disconfirming questions; added a note citing the debiasing/CDR-efficacy literature as the reason

for the removal; removed “Named bias specificity” as a listed strength in Section 11.1 and replaced it with a behavior-targeted framing.

- Reframed Section 3.1 (Represent/Bordage): stated explicitly that Bordage’s own attempts to demonstrate a causal effect of semantic-axis compression on diagnostic accuracy were unsuccessful; repositioned Step 2 as an externalization and communication tool rather than a demonstrated accuracy intervention.
- Expanded Section 1.4 (social/system factors) and folded it into the core conceptual model via Section 4.1’s design principles, Step 4, and Step 7.
- Rewrote the medico-legal bullet in Section 10: removed the framing of documented uncertainty as primarily a legal liability, and reframed the more significant barrier as cultural — clinician discomfort, particularly among junior clinicians, in explicitly acknowledging what they do not know — rather than treating documentation itself as the primary risk.
- Rewrote the institutional-buy-in bullet in Section 10: reframed the likely source of resistance away from clinicians feeling scrutinized by mandatory self-disconfirmation exercises, and toward experienced clinicians not seeing why they need to change a practice that has worked without an explicit model for years or decades.
- Rewrote Section 9: reframed rigor-vs-throughput tiering as a research question rather than a settled prescriptive table; added Section 9.1 distinguishing documentation burden from the case-volume/depth tradeoff raised in external feedback; added a corresponding Implementation Constraints bullet in Section 10.
- Shifted the paper’s stated identity toward the Reflect/Integrate stages as its primary defensible contribution: revised abstract framing, Section 3.4 emphasis, and Section 13.
- Added a named limitation in Section 11.2 addressing the foundational objection that procedural reasoning frameworks may not be the right unit of analysis for clinical reasoning at all.
- Added this Changelog, plus Reading Notes, Pilot Ideas, and Failure Logs appendices.
- Feedback throughout this appendix and the Failure Log (Appendix C) is unattributed to individual reviewers, since the outreach list includes the people whose feedback is described, and any letter-coding scheme would not stop them from recognizing each other’s critiques. The substance of every critique and how the paper responded to it is preserved.
- Narrowed the near-miss/comparison question in Section 8.1 from a claim about “unstructured reasoning” in general, which the single-arm pilot design in Section 8.2 cannot support, to a within-rater comparison against each rater’s own initial, pre-ARM impression of the case. Added a corresponding bullet to Section 8.2 instructing raters to record that initial impression before applying the ARM steps, so the narrowed question in 8.1 has something to compare against.
- Rewrote Step 5 (Section 5) and the corresponding row in the framework overview table (Section 4.2): test prioritization is no longer framed around how many competing diagnoses a result can move at once. A test’s value is now framed around its expected impact on the leading management decision, explicitly including the case where the highest-value test is one that decisively settles a single high-consequence hypothesis, rather than one that shifts probability across several.

- Rewrote Section 7, different example. Replaced a jargon-heavy anemia case with a sore-throat case, while preserving the bias risk, the disconfirming test, and the real-stakes teaching point.

Appendix B: Pilot Ideas

Ideas beyond the core case-series pilot in Section 8, generated in response to reviewer feedback. These are not commitments, they are candidates for a future pilot-design pass.

B.1 Volume-vs-Depth Pilot (responds to the case-volume opportunity-cost point, Section 9.1)

- Design sketch: compare two matched groups of learners over a fixed rotation block, one applying full ARM documentation to a subset of cases, one seeing an unconstrained case volume with no structured documentation, and track both reasoning-quality measures and total cases seen.
- Open question to resolve before genuinely designing this: what is the right reasoning-quality outcome measure, given Section 11.2's outstanding disagreement about whether procedural frameworks are even the right lens?

B.2 Attack-Step Only Micro-Pilot

- Isolate just Step 4 (Attack) as a brief, low-burden intervention layered onto existing case presentations, since it is the step with the most direct antecedent (Croskerry) and the clearest single behavioral ask.
- Would generate a faster feasibility signal than the full eight-step pilot in Section 8, at the cost of not testing the Reflect/Integrate stages this paper now treats as its main contribution.

B.3 Tier Self-Classification Check (Extends Section 9.2)

- Give raters case vignettes with no tier pre-assigned; ask them to classify into Tier 1/2/3 independently; measure agreement before ever testing whether the tiers themselves are the right depth recommendations.

B.4 Reflect/Integrate Longitudinal Pilot

- Since Section 3.4 now identifies Reflect/Integrate as ARM's real contribution, a dedicated longitudinal pilot (tracking a small cohort's Reflect/Integrate entries across a full rotation or year) may be worth more than further iteration on the early steps, which largely restate Bordage, Kassirer, and Mamede/Schmidt.

Appendix C: Failure Logs

A running record of critiques received, what they revealed, and how (or whether) this draft responded. Kept separately from the Changelog because the point of this log is to preserve the critique itself, not just the resulting edit, including critiques not yet fully resolved.

Core Critique	Status
Reasoning-vs-knowledge framing is not well supported; knowledge deficits, not reasoning process, are the dominant driver of diagnostic error.	Addressed: interaction model adopted (Section 1.2), now supported by verified citations located via independent search, not only citations raised in feedback.
Step 2 problem-representation compression has not been shown to causally improve diagnostic accuracy.	Addressed: Section 3.1 reframes Represent as an externalization tool, states the negative causal finding explicitly.
“Name the biases” has no evidentiary support; dozens of biases exist and there is no evidence clinicians can identify which is operating.	Addressed: instruction removed from Step 4; reason cited with verified supporting citations, including a direct empirical test of bias-identification agreement (Zwaan et al., 2017), located via independent search (Section 5, Step 4 note).
Deeper integration opportunity lies in what happens after the moment of reasoning: next steps, ongoing reflection, adaptive expertise.	Addressed: Section 3.4 and Section 13 now foreground Reflect/Integrate as the paper’s core contribution.
Koufidis’ work on social/situational influences on reasoning should be built in more explicitly.	Addressed: full citations located via independent search (Koufidis et al., 2021; Koufidis et al., 2022) and added to Section 1.4 and the reference list.
Rigor-vs-throughput tiering (Section 9) may be better framed as its own research question than as a proposed fixed balance.	Addressed: Section 9.2 reframes tiering as a research question; Appendix C.3 proposes a check.
Elaborate 8-step approach trades off against case volume, which is itself how System-1 pattern recognition is built.	Addressed: new Section 9.1 names this as a distinct cost from documentation burden; Appendix C.1 proposes a dedicated pilot.
Reference list is dated relative to how far clinical reasoning has moved; CDR/debiasing emphasis is now largely out of favor.	Addressed: independent literature search completed across two passes. Thirteen previously flagged citations added with full bibliographic detail; two current 2025–2026 sources added for currency.

Emphasized Mamede’s deliberate reflection literature, Koufidis’ work on social/situational factors, and the Reflect/Integrate stages as the paper’s most defensible original contribution.	Addressed: see the corresponding rows above; several of these points were raised independently by more than one reviewer.
Whether ARM (a prospective model) complements or competes with retrospective cognitive-error-analysis frameworks published in the journal <i>Diagnosis</i> .	Addressed: new Section 2.8 states this explicitly and now cites the specific curriculum discussed, Klimpl & Staudinger’s “Rooted in Reasoning” (2025), rather than describing it generically; the two are complementary, and that curriculum’s own module sequence (prospective foundations before a final retrospective module) illustrates the point directly.
Recommended expanding the positioning discussion beyond individual cognitive models to include structured/deliberate reflection, and clinical-reasoning teaching and assessment tools such as SNAPPS and the Assessment of Reasoning Tool (ART); engaging with this work would help define more precisely what ARM adds.	Addressed: SNAPPS and ART now have a full comparative treatment in Section 2.9: individual subsections for each tool and an honest statement of contribution (2.9.3), comparable in structure to the treatment given to Bordage, Kassirer, Croskerry, and Mamede/Schmidt in Section 3.
If a pilot goal is to determine whether ARM surfaces diagnoses or reasoning errors that unstructured reasoning would miss, the proposed design needs some form of comparison condition; as written, every rater uses ARM, so that question cannot be answered.	Addressed by narrowing: the question in Section 8.1 now asks whether ARM surfaces something the rater’s own initial, pre-ARM impression would have missed, a within-rater comparison the single-arm design can actually support. Section 8.2 adds a bullet instructing raters to record that initial impression before starting the ARM steps.
Pushed back on the Step 5 claim that a test’s value depends on how many competing diagnoses it can shift at once; the highest-value test is often one that settles a single consequential hypothesis.	Addressed: Step 5 (Section 5) and the framework overview table (Section 4.2) rewritten around expected impact on the leading management decision rather than breadth of hypotheses moved, a test that decisively settles one high-consequence hypothesis is now explicitly framed as potentially the highest-value option.
Recommended not overemphasizing medico-legal risk around documenting diagnostic uncertainty; in practice, documented uncertainty is often useful to downstream clinicians, and the larger barrier is cultural, particularly junior clinicians’ discomfort explicitly acknowledging what they don’t know.	Addressed: Section 10’s documentation bullet was rewritten to lead with the clinical value of documented uncertainty and reframe cultural discomfort, not legal exposure, as the primary barrier.
In Section 10, considered adding that since clinical reasoning is not universally taught, clinicians may resist not because they feel scrutinized but because they’ve practiced fine for years or decades without an explicit model and don’t see the need to change.	Addressed: Section 10’s institutional-buy-in bullet was rewritten around this reframing.

Raised concern that the terms “adversarial” and “attack”, even though the target is one’s own hypothesis, not another clinician, can carry an interpersonal connotation in medicine that could alienate some of the clinicians the framework is meant to reach; framed as a question of framing and implementation, not the underlying concept.	Considered and retained, not resolved by softening the language. The position taken in this revision: the terminology is not decorative. “Attack” specifies the cognitive posture the step requires, the clinician’s job in Step 4 is to actively try to break their own leading diagnosis, not to gently double-check it, and a softer term risks losing that behavioral precision. This is a defense of the mechanism, not of any particular implementation of it; the framing/implementation concern itself remains open and is not dismissed.
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